



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to P.M FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

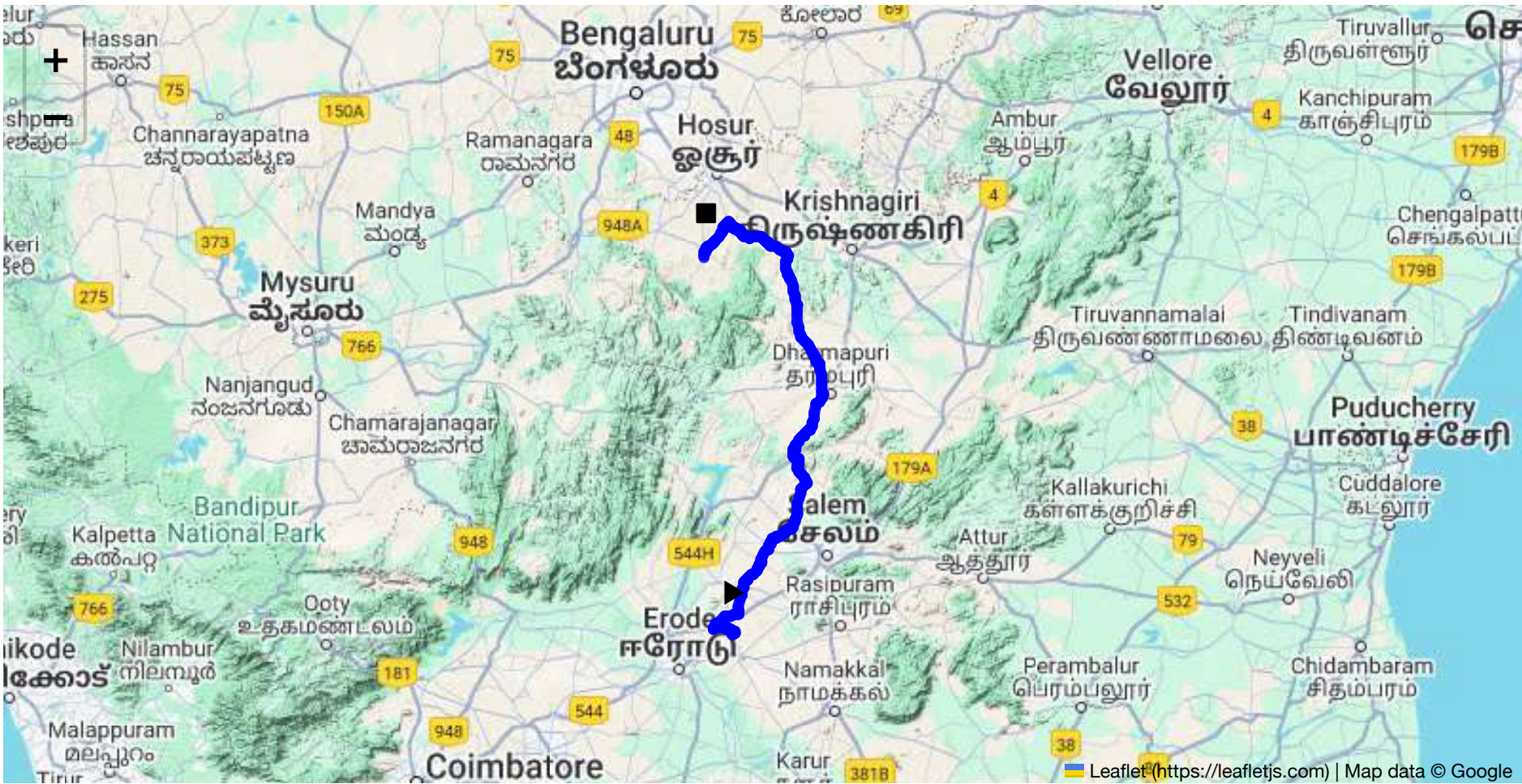
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 187.13 km
Estimated Duration: 3.9 hours
Adjusted Duration (Heavy Vehicle): 4.9 hours

Start: (11.4381, 77.8734)
End: (12.50901, 77.786567)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route spans approximately 187.13 kilometers from Sangagiri to Denkanikottai, passing through several key points including Bhavani Main Road, Ayveli, Erumaipatti, Salem, and Pillari Agraharam. It covers a mix of national and state highways, connecting rural regions with urban centers.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The monsoon season (October to December) may present hazards due to heavy rainfall, causing waterlogging and reduced visibility. Summers (March to June) can be extremely hot, potentially affecting vehicle performance, while the winter months (January to February) are moderate with minimal weather-related disruptions.

3. Analysis of Traffic Patterns

Salem, being a significant urban hub on this route, typically experiences heavy traffic during peak hours (8 AM - 10 AM and 5 PM - 8 PM). The stretch from Salem to Denkanikottai on SH 86 is prone to congestion due to ongoing roadworks and narrow lanes. Additionally, rural roads might encounter slow-moving agricultural vehicles.

4. Assessment of Road Quality and Infrastructure

The national and state highways in this region are generally in good condition, with paved roads and proper signage. However, some rural stretches might have uneven surfaces and potholes, particularly post-monsoon. Street lighting could be insufficient in rural areas, and caution is advised in hilly terrains near Denkanikottai.

5. Suggestions for Alternative Routes for Emergencies

In case of roadblocks or severe weather along the main route, consider using the NH44 towards Hosur and then proceeding to Denkanikottai. This can add distance but provides a more consistent highway infrastructure and better emergency assistance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to regulations such as the display of appropriate placards, sticking to designated routes, and ensuring the presence of necessary safety equipment. Checkpoint compliance near urban areas is strict, particularly in Salem.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been incidents involving overturning of heavy vehicles on sharp turns and inclines near Denkanikottai. Reports of leaks involving hazardous materials stress the importance of regular vehicle maintenance and periodic checks of safety equipment.

8. Environmental Considerations and Sensitive Areas

The route traverses environmentally sensitive forests near Denkanikottai. Drivers should respect local biodiversity and adhere to speed limits and noise regulations. Avoiding disposal of waste in these areas is crucial.

9. Analysis of Communication Coverage

The overall communication coverage along the route is good, but there may be potential dead zones in rural areas, particularly near Erumaipatti and forested stretches near Denkanikottai.

10. Estimated Emergency Response Times

Urban areas such as Salem have quick response times (15-30 minutes) due to proximity to services. In contrast, rural areas may experience delays (up to 60 minutes) depending on the specific location and availability of emergency services.

11. Overall Summary of Risk Assessment

This route is moderately risky due to potential weather impacts during the monsoon, urban traffic congestion, and rural road conditions. Emergency response preparedness is essential, particularly when transporting hazardous materials. Alternate routes and proactive maintenance checks are recommended to mitigate risks, alongside compliance with all transport regulations and environmental considerations.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	12.60628, 77.86014	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	Medium	11.43995, 77.87431	30 KM/Hr
3	Turn	High	11.44107, 77.87528	15 KM/Hr
4	Turn	High	11.44900, 77.87406	15 KM/Hr
5	Turn	Medium	11.45353, 77.85697	30 KM/Hr
6	Turn	Medium	11.45781, 77.85203	30 KM/Hr
7	Turn	High	11.46307, 77.84941	15 KM/Hr
8	Turn	Medium	11.45546, 77.81800	30 KM/Hr
9	Blind Spot	Blind Spot	11.45605, 77.81683	10 KM/Hr
10	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
11	Turn	High	11.49469, 77.88488	15 KM/Hr
12	Turn	Medium	11.57669, 77.90614	30 KM/Hr
13	Turn	High	11.59253, 77.90690	15 KM/Hr
14	Turn	Medium	11.63848, 77.94825	30 KM/Hr
15	Turn	Medium	11.64089, 77.95105	30 KM/Hr
16	Turn	Medium	11.64312, 77.95187	30 KM/Hr
17	Turn	Medium	11.64558, 77.95108	30 KM/Hr
18	Turn	Medium	11.67883, 77.96865	30 KM/Hr
19	Turn	Medium	11.70279, 77.97913	30 KM/Hr
20	Turn	Medium	11.71386, 77.99552	30 KM/Hr
21	Turn	Medium	11.71735, 78.00019	30 KM/Hr
22	Turn	Medium	11.71972, 78.01202	30 KM/Hr
23	Turn	Medium	11.72765, 78.02291	30 KM/Hr
24	Turn	High	11.73785, 78.03496	15 KM/Hr
25	Turn	High	11.73653, 78.03886	15 KM/Hr
26	Turn	High	11.74252, 78.04705	15 KM/Hr
27	Turn	Medium	11.77998, 78.05786	30 KM/Hr
28	Turn	Medium	11.86915, 78.08382	30 KM/Hr
29	Turn	Medium	11.87998, 78.08170	30 KM/Hr
30	Turn	Medium	11.91266, 78.05852	30 KM/Hr
31	Turn	Medium	11.92653, 78.06517	30 KM/Hr
32	Turn	Medium	11.93196, 78.06473	30 KM/Hr
33	Turn	Medium	11.93473, 78.06268	30 KM/Hr
34	Turn	Medium	11.93746, 78.05331	30 KM/Hr
35	Turn	Medium	11.95531, 78.04698	30 KM/Hr
36	Turn	High	11.95854, 78.04990	15 KM/Hr
37	Turn	High	11.96015, 78.04945	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.96266, 78.05115	30 KM/Hr
39	Turn	High	11.96433, 78.05113	15 KM/Hr
40	Turn	High	11.96542, 78.04938	15 KM/Hr
41	Turn	Medium	12.12036, 78.13284	30 KM/Hr
42	Blind Spot	Blind Spot	12.51393, 78.03186	10 KM/Hr
43	Turn	Medium	12.51379, 78.02933	30 KM/Hr
44	Turn	Medium	12.51624, 78.02577	30 KM/Hr
45	Turn	Medium	12.52269, 78.02290	30 KM/Hr
46	Turn	Medium	12.52651, 78.01023	30 KM/Hr
47	Turn	High	12.55108, 77.97248	15 KM/Hr
48	Turn	Medium	12.55286, 77.96596	30 KM/Hr
49	Turn	Medium	12.56524, 77.95283	30 KM/Hr
50	Turn	Medium	12.56246, 77.92215	30 KM/Hr
51	Turn	Medium	12.57304, 77.89875	30 KM/Hr
52	Turn	Medium	12.58211, 77.89398	30 KM/Hr
53	Turn	Medium	12.58409, 77.88837	30 KM/Hr
54	Turn	High	12.58358, 77.88760	15 KM/Hr
55	Turn	Medium	12.59952, 77.86311	30 KM/Hr
56	Turn	High	12.60497, 77.85468	15 KM/Hr
57	Turn	Medium	12.59990, 77.85037	30 KM/Hr
58	Turn	Medium	12.59351, 77.84812	30 KM/Hr
59	Turn	Medium	12.59167, 77.84620	30 KM/Hr
60	Turn	Medium	12.52347, 77.79152	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
4	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
9	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
11	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
13	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
22	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
24	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
25	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
26	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
27	hospital	Government Hospital, Jakkasamudram	12.3937246, 78.0522641	30 km/h	Medium
29	clinic	Kowthalam HSC	12.5676496, 77.8312573	30 km/h	Medium
31	hospital	Mythri Hospital, Krishnagiri	12.5264111, 77.7897745	30 km/h	Medium
32	police	Denkanikottai Police Station	12.5208971, 77.7903372	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
12	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
21	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
23	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
28	marketplace	Rayakottai tomato market	12.5237469, 78.0220377	30 km/h	Medium
30	school	St. Jopseph School	12.5339473, 77.795134	30 km/h	Medium
33	school	Govt. Boys Hr. Sec School	12.5158402, 77.787696	30 km/h	Medium
34	school	Holy Cross Hr. Sec. School	12.5114655, 77.7867735	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.60628, 77.86014



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44107, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45546, 77.81800



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57669, 77.90614



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.59253, 77.90690



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63848, 77.94825



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64089, 77.95105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64312, 77.95187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64558, 77.95108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67883, 77.96865



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70279, 77.97913



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71386, 77.99552



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71735, 78.00019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71972, 78.01202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72765, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73785, 78.03496



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73653, 78.03886



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74252, 78.04705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77998, 78.05786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86915, 78.08382



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87998, 78.08170



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.91266, 78.05852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92653, 78.06517



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93196, 78.06473



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93473, 78.06268



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96266, 78.05115



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96433, 78.05113



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96542, 78.04938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12036, 78.13284



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.51393, 78.03186



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51379, 78.02933



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.51624, 78.02577



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52269, 78.02290



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.52651, 78.01023



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.55108, 77.97248



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.55286, 77.96596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56524, 77.95283



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.56246, 77.92215



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.57304, 77.89875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58211, 77.89398



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.58409, 77.88837



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.58358, 77.88760



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59952, 77.86311



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.60497, 77.85468



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59990, 77.85037



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59351, 77.84812



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59167, 77.84620



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.52347, 77.79152